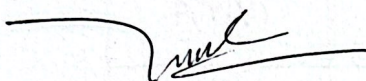


Final Examination

Date: 09/01/2026; Duration: 90 minutes

Offline, open-book, open-notes, and a calculator are permitted.

Laptops/cell phones are not allowed.

SUBJECT: Artificial Intelligence (ID: IT159IU)	
Approval by the School of Computer Science and Engineering Signature 	Lecturer: Nguyen Trung Ky Signature 
Full name: <i>Trần Thanh Tùng</i>	Full name: Nguyen Trung Ky
Proctor 1 Signature 	Proctor 2 Signature
Full name:	Full name:
STUDENT INFO	
Student name:	
Student ID:	

INSTRUCTIONS: The total points are 100 (equivalent to 40% of the course)

1. *Purpose:*

- Test your knowledge in Neural Network (CLO3)
- Test your knowledge and skill in Support Vector Machine (CLO3)
- Evaluate your skill in analysis and design in Neural Network (CLO3)
- Test your knowledge in Reasoning Under Uncertainty (CLO4)
- Evaluate your skill in analysis and design in Reasoning Under Uncertainty (CLO4)

2. *Requirements:*

- Read each question carefully and answer it following the requirements:
- Write the answers and draw models CLEAN and TIDY directly in the answer sheet.

Q1. [40 pts] Reasoning Under Uncertainty

1. [20 pts] Consider the following table of data.

A	1	0	1	2	0	1	2	1	2	0
B	0	2	2	1	2	2	1	0	0	1
Y	-	+	-	-	+	-	+	+	+	-

Find the following quantities. You can leave your answers as fractions.

- [5 pts] $P(Y = +) =$
- [5 pts] $P(A = 0 | Y = +) =$
- [5 pts] $P(A = 1 | Y = -) =$
- [5 pts] $P(B = 2 | Y = +) =$

2. [20 pts] We use a Naïve Bayes classifier to differentiate between Pacman and Ghosts, trained on the following:

F_1	F_2	Y
0	1	Ghost
1	0	Ghost
0	0	Pac
1	1	Pac

Assume that the distributions generated from these samples perfectly estimate the CP Ts. Given features f_1, f_2 we predict $\hat{Y} \in \{\text{Ghost}, \text{Pacman}\}$ using the Naïve Bayes decision rule. If $P(Y = \text{Ghost} | F_1 = f_1, F_2 = f_2) = 0.5$, assign $\hat{Y}$ based on flipping a fair coin.

a. [5 pts] Compute the table $P(\hat{Y} | Y)$ below?

Where value $P(\hat{Y} = \text{Ghost} | Y = \text{Ghost})$ is the probability of correctly classifying a Ghost, while $P(\hat{Y} = \text{Pacman} | Y = \text{Ghost})$ is the probability of confusing a Ghost for a Pacman.

Truth	predict	
	$P(\hat{Y} Y)$	
	$\hat{Y} = \text{Ghost}$	$\hat{Y} = \text{Pacman}$
	Y = Ghost	
	Y = Pacman	

For each modification below, recompute table $P(\hat{Y} | Y)$. The modification for each part are separate and do not accumulate.

- b. [5 pts] Add extra feature $F_3 = F_1 + F_2$, and modify the Naive Bayes classifier appropriately?

$P(\hat{Y} Y)$	$\hat{Y} = \text{Ghost}$	$\hat{Y} = \text{Pacman}$
$Y = \text{Ghost}$		
$Y = \text{Pacman}$		

- c. [5 pts] Add extra feature $F_3 = F_1 \times F_2$, and modify the Naive Bayes classifier appropriately?

$P(\hat{Y} Y)$	$\hat{Y} = \text{Ghost}$	$\hat{Y} = \text{Pacman}$
$Y = \text{Ghost}$		
$Y = \text{Pacman}$		

- d. [5 pts] Add extra feature $F_3 = F_1 - F_2$, and modify the Naive Bayes classifier appropriately?

$P(\hat{Y} Y)$	$\hat{Y} = \text{Ghost}$	$\hat{Y} = \text{Pacman}$
$Y = \text{Ghost}$		
$Y = \text{Pacman}$		

Q2. [30 pts] Neural Network

1. [5 pts] Please show the following statements is True or False.

If while running the perceptron algorithm we make one pass through the data and make no classification mistakes, the algorithm has converged.

2. [25 pts] We would like to use a perceptron to train a classifier for the following datasets with two features, A and B, and binary class +1 or -1.
Consider the following labeled training data:

Index	A	B	Class
1.	0	0	1
2.	3	3	-1
3.	1	1	1
4.	3	0	-1
5.	-1	1	1
6.	0	-3	-1
7.	-1	-1	1
8.	-3	0	-1

- a. [5 pts] Please show each of the following statements is True or False.
- This data is linearly separable
 - For all valid kernels, there exists a dataset with at least one point from each class that is linearly separable under that kernel.
- b. [5 pts] Our two perceptron weights have been initialized to $w_1 = 3$ and $w_2 = -3$. After processing what is the first misclassified point with the perceptron algorithm?
- c. [5 pts] What will be the updated values for these weights?
- d. [5 pts] What is the second misclassified point?
- e. [5 pts] For the perceptron algorithm, what is the weight vector after the second weight update?

Q3. [30 pts] Support Vector Machine

Suppose we have three one-dimensional data points as follows:

$$x_1 = 0, y_1 = 1; x_2 = 1, y_2 = -1; x_3 = 2, y_3 = 1.$$

We use the polynomial kernel of degree 2

$$K(x_i, x_j) = (x_i x_j + 1)^2$$

Assume a large C so the problem is effectively hard-margin and all optimal α_i are interior:

$0 < \alpha_i < C$. Under that assumption the KKT equalities hold for every training point.

[30 pts] Compute $\alpha_1, \alpha_2, \alpha_3$ and b by solving the dual linear system.

– GOOD LUCK –